ModelIt!

Teacher's Guide

How Can We Protect Against Natural Hazards?

Engineering Solutions for Dangerous Events

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-ESS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-ESS3-1

Make a claim about the merit of a design solution that reduces the impacts of a weather-related hazard.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students make claims about the merit of design solutions that reduce the impact of weather-related hazards, supporting their arguments with evidence from model simulations.
Disciplinary Core Idea	**ESS3.B Natural Hazards** — A variety of natural hazards result from natural processes. Humans cannot eliminate natural hazards but can take steps to reduce their impacts.
Crosscutting Concept	**Cause and Effect** — Students investigate how engineering design features (cause) reduce the damage from natural hazards (effect), and evaluate which designs are most effective against different hazard strengths.

CAST Testing Alignment

MULTIPLE CHOICE:

Two houses are in a flood zone. House A is built on stilts 3 feet above the ground. House B sits flat on the ground. A flood sends 2 feet of water through the area. Which house has less damage, and why?

CONSTRUCTED RESPONSE:

Design a solution to protect a school from a natural hazard in your area. Explain how your design reduces damage. Use the words 'hazard,' 'protection,' and 'damage reduction' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Natural hazards have threatened humans throughout history. While we cannot stop earthquakes, floods, or tornadoes from happening, engineers have developed remarkable solutions to protect people and reduce damage.

Types of Natural Hazards

Natural hazards include weather events (floods, tornadoes, hurricanes, blizzards, droughts), geological events (earthquakes, volcanic eruptions, landslides, tsunamis), and wildfires. Each type of hazard causes damage in different ways — floods push water into buildings, tornadoes rip structures apart with wind, and earthquakes shake buildings until they collapse. Understanding HOW each hazard causes damage is the first step in designing protection against it.

Engineering Solutions

Engineers have developed many solutions to reduce hazard damage. Levees and floodwalls hold back floodwater. Storm shelters protect people from tornado winds. Earthquake-resistant buildings use flexible steel frames that sway without breaking. Hurricane shutters protect windows from flying debris. Seawalls shield coastlines from storm surges. Fire-resistant materials and defensible space around homes reduce wildfire damage. Each solution is specifically designed to counter the way that particular hazard causes harm.

Reducing Impact, Not Eliminating Risk

It is important to understand that engineering can REDUCE hazard impacts but never eliminate them completely. A levee can hold back a 20-foot flood but may fail in a 30-foot flood. A tornado shelter protects against most tornadoes but could be damaged by the most extreme ones. Engineers design for the most likely hazard levels, using historical data to determine how strong protection needs to be. Building stronger costs more money, so communities must balance protection level with cost.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify hazard strength and protection design as components — hazard strength is external (we cannot control it) while protection design is also external (we choose how to build it). Damage reduction is the internal outcome.

E **Establish Relationships**

Students determine that stronger protection design leads to more damage reduction, while stronger hazards overwhelm weaker protections. The relationship between protection and reduction is positive.

V **Visualize & Model**

Students build a model showing how the interaction between hazard strength and protection design determines how much damage is prevented.

E **Evaluate Outcomes**

Students run multiple scenarios combining different hazard strengths with different protection levels to find the best engineering solutions.

R **Revise & Extend**

Students add warning time or building materials to explore how early evacuation and material choice affect overall damage reduction.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What would you do if a big flood was heading straight for your neighborhood? How would you protect your home?"*

Watch For

Show photos of natural hazards and engineering solutions side by side. Ask students what they notice about how people prepare.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we become engineers! We are going to figure out how to protect people from dangerous natural events!"*

Watch For

Read objectives together. Introduce vocabulary — show examples of natural hazards and ask which ones students have heard about.

Slide 3: The Big Question

3 min

Say: *"We cannot stop floods, tornadoes, or earthquakes. But CAN we build things that keep people safe?"*

Watch For

Show before-and-after photos of a flood — protected buildings vs. unprotected. Ask: What made the difference?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model that tests different protection designs against different hazard strengths!"*

Watch For

Preview each LEVER step. Explain that civil engineers use models to test designs before building them in real life.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of this protection system can we identify? What can we control, and what is the result?"*

Watch For

Guide sorting of three components. Discuss that both hazard strength AND protection design are inputs, but we can only control the protection.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When we make protection stronger, does damage reduction go up or down? What about when the hazard gets stronger?"*

Watch For

Help students draw arrows showing that stronger protection increases damage reduction, while stronger hazards challenge the protection.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Engineer test time! Does strong protection save us from a big storm? Does weak protection help at all?"*

Watch For

Have students predict, then run each scenario. Compare results to see which combinations prevent the most damage.

Slide 8: What Did We Discover?

7 min

Say: *"Good engineering saves lives! We cannot stop natural hazards, but we CAN build smarter to reduce the damage."*

Watch For

Summarize discoveries. Show real examples of engineering solutions (levees, storm shelters, earthquake-proof buildings). Discuss what our community uses.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Hazard Strength, Protection Design
- **INTERNAL:** Damage Reduction

Relationships

- **Protection Design to Damage Reduction:** POSITIVE (+) — Stronger, better-designed protection reduces more damage. A house on 4-foot stilts prevents more flood damage than a house on 1-foot stilts. A reinforced concrete storm shelter provides more tornado protection than a wooden shed. Better engineering means less harm.
- **Hazard Strength to Damage Reduction:** POSITIVE (+) — Note: this relationship means that as hazard strength increases, the CHALLENGE to damage reduction increases. Very strong hazards can overwhelm even good protection. A Category 5 hurricane may damage even well-engineered buildings, while a Category 1 hurricane causes little damage to the same structures.

Key Discoveries

- Better-designed protection reduces more damage from natural hazards — engineering saves lives
- Even weak protection helps some, but strong hazards require strong engineering solutions
- No design can prevent ALL damage from the most extreme natural events — but good design greatly reduces harm
- Engineers study past disasters to design better protection for the future

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Technology can protect us from all natural hazards completely"	Even the best engineering cannot prevent ALL damage from extreme natural events. A massive earthquake, a Category 5 hurricane, or a once-in-a-thousand-year flood can overwhelm any human-made structure. Engineering reduces risk and saves lives, but it cannot eliminate all danger from nature's most powerful forces.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Natural hazards are completely unpredictable and random"	While we cannot predict the exact moment a hazard will occur, scientists use data and patterns to identify WHERE hazards are most likely and roughly WHEN certain hazards are more common. Hurricanes have a season (June-November). Earthquakes cluster along fault lines. Floods happen in low-lying areas near rivers. This knowledge helps engineers build protection where it is needed most.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The best solution is always to just build things stronger"	Building stronger is one approach, but it is not always the best or only solution. Sometimes the smartest solution is to NOT build in dangerous areas at all, or to design buildings that move with the hazard (like flexible earthquake-resistant frames) instead of trying to resist it. Warning systems, evacuation plans, and community preparedness are just as important as physical structures.